Segment Specialization in Active-Dendrite Continual Learning: An Audit, an Artifact, and a Causal Repair

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Abstract

Active-dendrite architectures route each input through a competition among dendritic segments, one of which is trained per task; usage-localized elastic weight consolidation (DeLP) protects what has been learned by scaling per-neutron Fisher anchors with segment usage. Both mechanisms rest on an implicit assumption that has, to our knowledge, never been measured: that the segment–task mapping constitutes a *specialization* that protection can leverage at sub-neuron granularity. We audit this assumption on Permuted-MNIST and Permuted-Fashion and report three findings. First, a granularity ladder of anchors (plain, neuron-resolved, segment-resolved with attribution-tracked reference states) shows that the entire benefit of usage localization is realized at neuron level: segment-resolved anchors are statistically indistinguishable from neuron-resolved anchors across every combination of two datasets, two task counts (T=10, 20), and two elastic weights ($|\Delta\ell| \leq 0.24$, all n=10 paired tests n.s.), and remain so under FIFO anchor budgets ($|\Delta\ell| \leq 0.44$ across eight budget cells). Second, the audit explains why: task-dominant gating is *absent* — among active neurons the probabil-

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ity that a neuron's dominant segment is its own task's segment starts each phase at chance and is pushed *below* chance by that phase's own training (T=20: 0.05 $\rightarrow$ 0.01), and Fisher mass is diffuse across segments (top-1 share 11–18%). The audit also uncovers a measurement artifact relevant to the active-dendrite literature: under production sampling budgets 56–63% of neurons are never activated, and naive argmax instrumentation over all neurons fabricates a strong "segment-0 dominance" ($\approx$ 60%) that vanishes under active-only filtering. Third, because the missing specialization is load-bearing, restoring it *causally* improves the retention–plasticity trade-off: a constant bias on the own-segment dendritic input raises the learning term in every cell of a $\{\text{dataset}\} \times \{\text{depth}\} \times \{\text{granularity}\}$ grid ($\Delta\ell > 0$ in 18/18 dose cells, significant in 16/18 at n=10 paired), with final accuracy improving in step (15/18 simultaneously significant, and every grid cell significant on both terms at some dose; $\Delta\ell$ up to +4.6, $t(9)$=8.1); forgetting is reduced at T=10 (ΔBWT up to +1.7) and mildly traded back at T=20 (significant in 3/10 cells). Signed-drift instrumentation shows the network never counteracts the bias at the parameter level; the own-segment win rate saturates at 0.13–0.19 regardless of dose because any effective dose dominates the raw gate scale — partial restoration suffices. A companion mechanism probe (S4) then revises the reading: the repaired gate is *context-degenerate* (swapping the context one-hot changes 0.0% of predictions at b=10), and a gate-removal control reproduces the gain on MNIST, so the bias pays by removing the gate's dependence on context rather than by restoring segment–task routing. Segment–task specialization in this architecture family is thus not a weak tendency but an actively violated assumption — and one whose absence the

repair compensates for rather than cures.

Keywords: continual learning, elastic weight consolidation, active dendrites, dendritic segments, catastrophic forgetting, interpretability audit

1. Introduction

Biological neurons integrate thousands of synaptic inputs across morphologically distinct dendritic compartments, and single-neuron studies have long argued that these compartments act as quasi-independent computational subunits [1, 2, 3]. An influential line of work has translated this principle into artificial networks: context signals select one of several dendritic *segments* per neuron, a winner-take-all competition among segments determines the nonlinear contribution to the soma, and *only the selected segment receives gradient* on a given task [4]. The architectural promise is a division of labor: if task k's computation flows predominantly through segment k, then tasks naturally occupy disjoint sub-network state, and interference between tasks has a structural bottleneck.

Continual-learning methods built on dendritic architectures inherit this promise, often silently. DeLP (usage-localized elastic weight consolidation) protects per-neuron parameter blocks with Fisher anchors scaled by how heavily each neuron was used in past tasks [5, 6]. Its protection unit is the neuron; the segment structure below the neuron is used only to define usage. A natural proposal — one we initially expected to succeed — is to push protection *down* to the segment level: if tasks really live in different segments, anchors that track each segment's own reference state should protect more precisely and drag less on future learning, echoing the granularity intuition

that finer protection localizes interference better.

This paper reports what actually happens when that assumption is tested rather than assumed. Our contributions are threefold, and each arose from the failure of the previous one:

1. **A granularity audit with a null result at segment level (Sec. 3).** We implement segment-resolved anchors with *attribution-tracked reference states* — each anchor tracks where its own segment's parameters drift — and compare a three-rung ladder (plain, neuron-resolved, segment-resolved) across two datasets, two depths, and two elastic weights with n=10 paired seeds. The entire benefit of usage localization is realized in the plain→neuron step ($\Delta\ell$ between -2.8 and -5.4, $|t(9)|$ up to 22); the neuron→segment step adds nothing ($|\Delta\ell| \leq 0.24$, all n.s.), under unbounded stacks and under FIFO anchor budgets alike. The null is mechanistically pinned, not merely observed: attribution is *self-limiting* (the anchor's own penalty pins the weights, so the tracked reference never moves) and segment-level parameter ownership lacks a separable carrier.
2. **A measurement artifact with literature-wide relevance (Sec. 4).** Auditing *why* the null occurs, we find that under production sampling budgets 56–63% of neurons are never activated, and that naive dominant-segment instrumentation (argmax over segment activations of *all* neurons) attributes every dead neuron to segment 0 — fabricating an apparent "segment-0 dominance" of $\approx$ 60% that vanishes under active-only filtering. Among active neurons the true picture is *anti-specialization*: each phase's own training pushes the own-segment

win rate from chance to below chance (T=20: 0.05 $\rightarrow$ 0.01), and anchor Fisher mass is diffuse across segments (top-1 share 11–18%). Any study that measures segment routing with naive argmax — including evaluations of segment specialization claims — is at risk of the same artifact.

3. **A causal repair (Sec. 5).** If missing specialization is load-bearing rather than cosmetic, restoring it should help. It does: adding a constant bias b to the own-segment dendritic input during training, evaluation, and anchor construction raises the learning term in every cell of a $\{\text{dataset}\} \times \{\text{depth}\} \times \{\text{granularity}\}$ grid — $\Delta\ell$ positive in 18/18 dose cells, significant in 16/18 (n=10 paired) — with final accuracy significant simultaneously in 15/18 and in every grid cell at some dose ($\Delta\ell$ up to +4.6 at T=20, $t(9)$=8.1); forgetting is *reduced* at T=10 (ΔBWT up to +1.7, $t(9)$ up to 10.7) and mildly traded back at T=20 (significant in 3/10 cells) (Sec. 5). The response is saturated and nontrivially mediated: the own-segment win rate saturates at 13–19% regardless of dose. Signed-drift measurement shows this is not adversarial compensation — the network never drifts against the bias — but a scale effect: the raw own gate spans $|u| \leq 0.32$, at least three times below any tested dose, so the bias forces the same training-time routing at every dose, so only a minority of neuron-tasks flip to the own-segment pathway — and that partial restoration already yields most of the gain.

Together the three results reframe segment specialization in this architecture family: not a weak tendency to be exploited, but an actively violated assumption that can be diagnosed with corrected instrumentation and par-

tially repaired with a one-parameter intervention. All experiments are on Permuted-MNIST and Permuted-Fashion at $T \in \{10, 20\}$ with paired-seed statistics; code, per-run records, and analysis scripts are public (Data Availability).

2. Background and setup

2.1. Active-dendrite network

We use the feed-forward active-dendrite network of Iyer et al. [4] in the protected variant established in our prior work [5]. A one-hidden-layer network of N_h neurons receives a one-hot task context $c_t \in \{0, 1\}^T$; each neuron j carries T dendritic segments, and segment k of neuron j computes $u_{j,k} = \sigma(w_{(j,k)}^\top x + b_{j,k} + v_k^\top c_t)$, where $v \in \mathbb{R}^{T \times (n_{seg} N_h)}$ are the context-to-dendrite weights — *column* k of v gates segment k for every neuron. The soma nonlinearity applies a k-winners-take-all (kWTA) selection over $h_j = f_j(x)\, \sigma(\max_k u_{j,k})$: the k largest somal pre-activities pass; the rest are set to zero. Crucially, task t's gradient updates are masked to the parameters of segment t ("protect_grads"): the architecture's design intent is that task t trains segment t.

2.2. DeLP: usage-localized EWC and the identity decomposition

DeLP [5] protects the feed-forward and output blocks with per-task anchors, each an elastic penalty $\frac{\lambda}{2} \sum_p F_p\, (W_p - \theta_p)^2$ whose Fisher term F is *usage-localized*: per-neuron usage $o_j \in [0, 1]$, the frequency with which neuron j is active under task t's data, scales the anchor rows of the hidden weights and columns of the output weights, so that rarely-used neurons are

protected lightly and remain available to future tasks. Writing $\bar{a}_T$ for final average accuracy, the decomposition

$$\bar{a}_T \;=\; \ell \;+\; \tfrac{T-1}{T}\,\mathrm{BWT}, \tag{1}$$

separates what each task achieves when it is learned (ℓ, the learning term: the mean over tasks of the accuracy on a task immediately after that task has been trained) from what earlier tasks retain (BWT ≤ 0); we report all three, with paired t-statistics over n=10 seeds sharing permutation streams. Where a table quotes an accuracy on the *final* task alone, it says so explicitly (Table 2).

2.3. Protocol

Benchmarks are Permuted-MNIST and Permuted-Fashion [7, 8] with $T \in \{10, 20\}$ tasks, one hidden layer of 2000 neurons, 15 epochs per task, Adam (lr 10^{-3}), Fisher/usage samples of 300/500 examples, and $\lambda \in \{1000, 5000\}$ calibrated in prior work. Seeds are paired across arms (identical permutation streams and init); all comparisons are paired t-tests with n=10, and we mark $|t| > 2$ as significant. All hypotheses in Secs. 3 and 5 were preregistered in writing before the corresponding data were collected.

3. Does segment-resolved protection help? A granularity ladder

3.1. Segment-resolved anchors with attribution-tracked references

A segment-resolved anchor must differ from a neuron-resolved one in more than bookkeeping: since $\sum_k o_{t,j,k} = o_{t,j}$ (segment usages partition neuron usage), splitting a neuron's single Fisher scalar across segments while sharing

one reference state θ leaves the penalty algebraically identical to neuron level. We therefore give each segment attribution of parameter drift: anchor a (task t, neuron j) maintains per-segment reference states $\theta^{(a,k)}$, and after each subsequent task t' the realized parameter displacement is partitioned across segments by the usage-driven attribution weights of t''s own anchors; the penalty charges anchor a only for the drift not explained by interlopers in its own segment. The design intends that an anchor no longer litigates, on shared weights, drift caused by tasks that never touched its segment. Segment-level Fisher mass $m_{t,j,k}$ and dominant-segment histograms are instrumented per anchor.

3.2. Result: the ladder is flat above neuron level

Table 1 shows the ladder. The plain→neuron step is large and universal: at λ=1000, T=10 MNIST the learning term gains +4.66 ($t(9)$=22.3); across all six cells of {dataset} × {depth} × {λ}, $\Delta\ell$ ranges from -2.78 to -5.44 in favor of neuron-level localization with $|t(9)| \in [10.4, 22.3]$. The neuron→segment step is empty on the learning axis: the largest deviation across all twelve pairings is $|\Delta\ell| = 0.24$, never significant (largest $|t| = 1.3$); BWT differences are within 0.43 points. Two of those BWT differences are nominally significant (effect sizes below 0.5 points, $|t| \leq 4.0$) and we report them rather than suppress them: they are marginal forgetting effects, not evidence for a granularity effect, since they sit on the stability side rather than the learning side that the pre-registered hypothesis was about.

Single-anchor references (Table 2) frame the arms; their λ values are the calibration carried over from our companion pipeline, not re-tuned per benchmark. Plain EWC buys ℓ at 3–5× the forgetting; SI (λ=100) is over-

stable (its ℓ_{final}=13.2 at T=10 shows it barely learns the later tasks at all, even though its mean learning term ℓ=40.0 is not unremarkable); MAS (λ=1) maximizes the learning term ($\ell \geq 85.7$, the highest of all arms) yet posts catastrophic BWT (-32.6 at T=20) — the regime-dependence of warrant documented in our companion work extends from Split-CIFAR-100 to the MLP setting. One comparison the table makes possible is worth stating explicitly: at T=10, the repaired DeLP arm (b=10; avg 68.7, BWT -9.9) dominates MAS (avg 66.6, BWT -21.2) on both summary terms — MAS buys its higher learning term with exactly the forgetting that the repair removes.

3.3. The flatness survives anchor budgets

Because segment-resolved anchors might pay off only when the anchor *stack* is bounded — fewer anchors plausibly sharpen the value of each — we crossed granularity with FIFO budgets $M \in \{2, 5\}$ at λ=1000 (preregistered: alive iff $\geq 4/6$ cells significant positive and none significantly worse on BWT). The result is the same null, tighter: across all eight $\{\text{dataset}\} \times \{M\} \times \{T\}$ cells the branch$-$neuron $\Delta\ell$ lies within ± 0.44 with no cell significant positive (one cell, Fashion M=2, T=10, is significantly *negative* at -0.33, $t(9) = -2.3$; Table 3). Budgets shift both granularities identically, consistent with the shared mechanism analysis below.

4. Why the ladder is flat: auditing segment–task specialization

The null in Sec. 3 would be easy to misread as "the segment machinery is redundant." The audit says something stronger: the specialization

that segment-resolved protection would need does not exist, and part of the evidence that appeared to show it was an artifact.

4.1. A measurement artifact: dead neurons fabricate dominance

Our first-pass instrumentation recorded, for every neuron, the argmax over its segment activations under each task's data. It showed segment 0 "dominating" 58–63% of neurons — a striking positional bias that initially looked like the failure of task-specific routing. The correction: under the production sampling budget (500 examples), 56–63% of neurons are *never activated*; for a dead neuron all segment activations are exactly zero, argmax breaks ties toward index 0, and the neuron is silently booked to segment 0 (Fig. 1). Filtering to active neurons collapses the apparent bias to chance ($\approx 0.10 = 1/T$ at T=10; untrained networks show no positional preference either, 0.096–0.111 over three seeds). The death itself is real and grows with the sequence: the inactive fraction jumps from 0.02 to 0.35 within the first phase of training and stabilizes near 0.6 (Fig. 2) — under kWTA, neurons pushed out of the winner set early are never re-selected by any later task. We flag this as a methodological warning for the active-dendrite literature: dominance statistics computed over all neurons conflate routing with mortality, and at production sampling budgets the conflation is an order of magnitude larger than the signal.

4.2. Anti-specialization among active neurons

With the artifact controlled, the true routing picture is the opposite of specialization. Define p_{own} as the probability that a neuron's dominant segment is the currently measured task's own segment, over active neurons. At

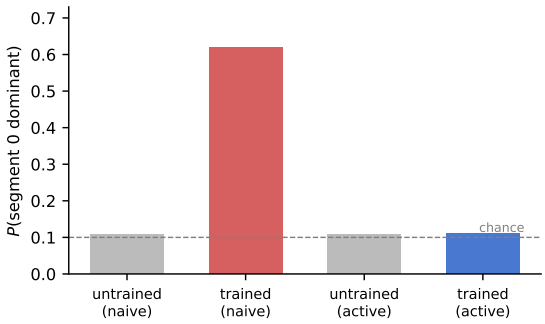


Figure 1: The dead-neuron artifact. Naive argmax over *all* neurons reports segment-0 dominance of $\approx 60\%$ after training; the same statistic over active neurons only, and both statistics on untrained networks, sit at chance ($1/T$).

the *start* of each phase, p_{own} sits at chance ($1/T$): nothing in the context-gated routing maps tasks to segments. Training within the phase then pushes it *below* chance — by the end of phase p, p_{own} drops to 0.058–0.085 at T=10 (chance 0.10) and, late in a T=20 sequence, to 0.005–0.02 against a chance of 0.05, i.e. five to ten times *below* random assignment (Fig. 3). The own-segment gating block does train — it moves by 6–16% of its norm per phase — yet this movement fails to convert into dominance: win rates still end each phase below chance. Segment–task alignment is not merely absent; its absence persists through active training of the very parameters that would encode it.

4.3. Fisher mass is diffuse across segments

Segment-resolved anchors also lack a substrate to hold onto: for trained anchors, the fraction of a neuron's Fisher mass attributable to its top segment is only 11–18% (mean over anchor–neuron pairs). Even a perfectly attributed segment-level penalty would spread nearly all of its protection

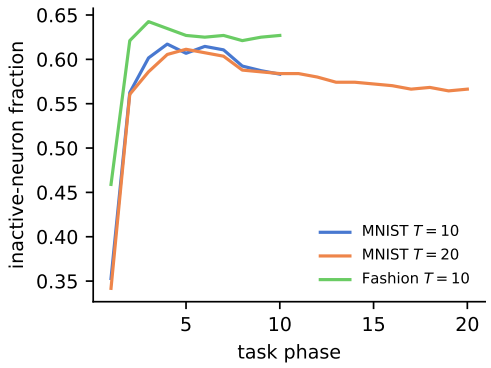


Figure 2: Inactive-neuron fraction across the task sequence (end-of-phase means). Neurons lost in early phases are never recovered; the active substrate available to routing is roughly halved by phase 2 and stabilizes near 40%.

across segments the anchor task never drove. This measurement closes the mechanistic account of the Sec. 3 null from two sides: attribution is self-limiting (an anchor's own penalty pins the shared weights it measures, so its tracked references barely move), and the per-segment mass that attribution would partition is diffuse to begin with.

5. Causal probe: restoring own-segment advantage

5.1. A one-parameter intervention

If the missing specialization is load-bearing, causally restoring it should improve the trade-off in Eq. (1). We add a constant bias b to the own-segment dendritic input: $u_{j,t} \mathrel{+}= b$ whenever the context selects task t. The bias is active in training, evaluation, and anchor construction (the Fisher measurement that drives usage-localized protection); the dominant-segment win-rate instrument deliberately reads the raw, bias-free gates, so that every

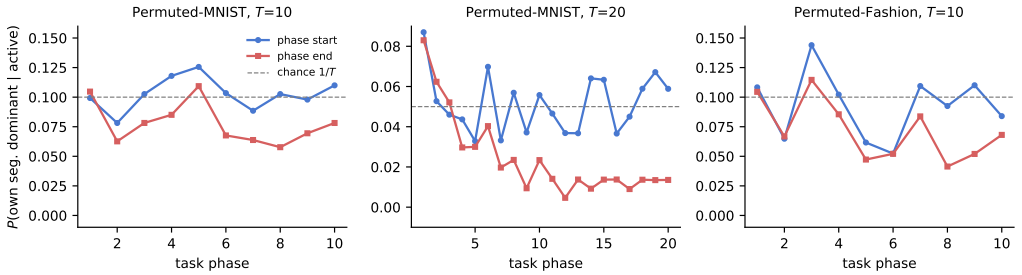


Figure 3: Anti-specialization over training. Within each phase, the own-segment win rate over active neurons starts at chance (dashed) and ends below it; the deficit deepens with sequence position (rightmost points of each curve), most severely at T=20.

routing statistic in this paper describes what the network itself encodes rather than what the bias forces (Sec. 5.3). The intervention is deliberately dumb — no learned parameters, no schedule, no per-neuron structure — so that any effect is attributable to the routing prior alone. Doses $b \in \{0, 1, 3, 10\}$; all comparisons paired against b=0 on identical seeds.

5.2. Result: the learning term and accuracy improve everywhere

Tables 4 and 5 give the dose–response; Fig. 4 plots it. The full grid is 2 datasets × 2 depths × 2 granularities × 3 doses = 24 cells; 22 of them are reported here (the two T=20 branch cells at b=1 were not collected), and across the 18 cells we do report, $\Delta\ell$ is positive in every cell and significant in 16/18; ℓ and final accuracy are simultaneously significant in 15/18, and every grid cell has at least one dose significant on both terms. The effect grows with depth: at T=20 MNIST, b=10 lifts ℓ by +4.56 ($t(9)$=8.1) and final accuracy by +3.81 ($t(9)$=10.6) over the already usage-localized baseline; the T=20 learning curve is monotone in dose ($\Delta\ell$ +2.14 / +3.72 / +4.56 for b=1/3/10, all $t \geq 3.9$), and b=3 alone already captures 81% of the b=10 effect

there (81–113% across the six dataset/depth/granularity arms we ran at all three doses). Forgetting improves broadly at T=10 (ΔBWT up to +1.65; significantly positive in 6/8 cells, and in all four MNIST cells), and partially trades back at T=20, where 3/10 cells show a significant mild BWT cost (-0.66 to -0.95) and the remaining seven are statistically indistinguishable from baseline.

Two robustness notes. First, the 18/18 sign uniformity of $\Delta\ell$ is itself the strongest guard against multiplicity: a sign test gives $p = 2^{-18}$, orders of magnitude below any family-wise correction over the grid. Second, the intervention also reduces neuron mortality — measured on the raw, bias-free network, the mean dead fraction drops by 3.9pp (MNIST) and 2.3pp (Fashion) at T=10 — and per-run $\Delta\ell$ correlates with the mortality reduction ($r = -0.49$ and -0.68; both correlations are computed on the ten paired seeds of the T=10 neuron arm at b=10, the same cell in which the dead-fraction shift is quoted). We read this as one causal chain rather than a confound: raising the own-segment gate raises sigmoid(u) during training, which increases kWTA survival, so forcing the routing prior is also what keeps neurons alive; the routing and capacity channels are two faces of a single intervention, not competing explanations.

5.3. The intervention is absorbed, not opposed

The mediation of the effect is as informative as its size. The bias does *not* flip measured routing: the own-segment win rate rises only from ≈ 0.08 to a saturated 0.13–0.19, and is nearly insensitive to a tenfold dose increase (b=1: 0.19; b=3: 0.17; b=10: 0.16). The dose-invariance has a scale explanation rather than an adversarial one. The raw own-segment gate that the win-rate

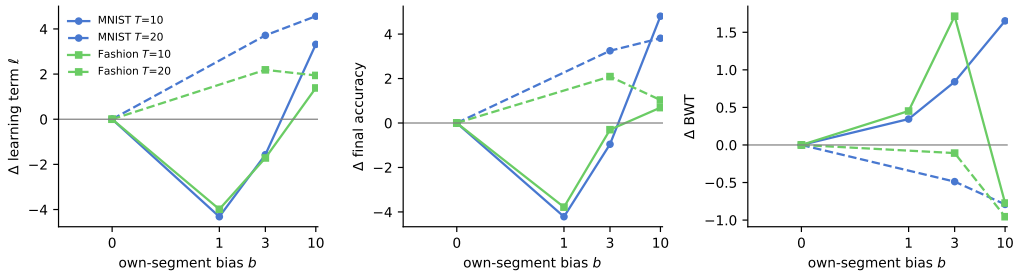


Figure 4: Dose–response of the own-segment bias (neuron-resolved arm, paired means vs. b=0). The learning term (left) and final accuracy (middle) improve at every dose, dataset, and depth; BWT (right) improves at T=10 and is mildly negative at T=20 on MNIST. Note the saturating shape: b=3 alone already captures most of the b=10 effect (81–113% across datasets and depths).

instrument reads stays within $|u| \leq 0.32$ across all seeds and phases (seed means within ± 0.16; Sec. 5.5); every tested dose is at least three times this range, so the bias, not the raw gate, determines the own segment's training-time advantage. The resulting parameter-level trace is dose-invariant, and hence so is the measured win rate. Signed-drift measurement (Sec. 5.5) confirms that the network never counteracts the bias at the parameter level: the own gating column drifts *positive*, not negative, under the imposed bias. Two conclusions follow. First, the improvement is not an artifact of forcing hard routing — the measured routing solution survives essentially intact. Second, only a minority of neurons needs to respect the segment–task prior for the retention–plasticity balance to shift significantly: specialization operates as a bottleneck resource, and partial restoration is sufficient.

A caveat on the mechanism is due, and a companion study (S4, [9]) supplies it. The dose-invariant saturation above is consistent with two very

different pictures: the bias nudges a minority of neurons toward their own segment (partial restoration), or it saturates the winning segment's sigmoid until the gate stops depending on the context at all (bypass). A context-swap probe distinguishes them: it shows the repaired gate is *context-degenerate* — after b=10 the fraction of predictions changed by swapping the context one-hot is exactly 0.0% (versus at most 6.2% at b=0) — and a degenerate-control ablation, in which the dendritic gate is removed entirely ($h = \mathrm{kwta}(t)$), reproduces the repair's full benefit on MNIST (73.22 vs. 72.63, n.s.). On that dataset, therefore, what this section measures is a repair that pays by *removing* the gate's dependence on the context, not by restoring segment–task routing. The retention and learning numbers reported here are unaffected by which reading is correct — but the reading matters, and we adopt the bypass interpretation for this family rather than the partial-restoration one.

5.4. Task relevance: where the repair pays off

Is the repair a *routing prior* (should help most on tasks the network already understands) or a *capacity channel* (should lift all tasks uniformly)? We test this directly with a task-relevance gradient (Suppl. S3): ten interleaved phases in which two in-domain digit/parity tasks (high relevance), two rotated-count tasks (medium), and six random-label tasks (zero relevance) are learned in sequence, with the bias varied over $b \in \{0, 1, 3, 10\}$ (n=10 paired seeds, both granularities). Per-task diagnostic accuracy (ℓ, each task measured against its own training labels) is compared across the three groups.

Three findings. **First**, zero-relevance tasks are unaffected by the bias: their per-task accuracy sits at chance ($\approx 10\%$) at every dose, and the dose–response on this group is flat (9.99–10.07 from b=0 to b=10). The repair adds

no capacity where there is no structure to exploit. **Second**, the gap between the high-relevance group ($\ell \approx 85$) and the medium group ($\ell \approx 71$) *narrows* monotonically as the dose grows (+13.8 at b=0 to +11.2 at b=10): the repair compresses task-to-task variance rather than lifting the floor, consistent with a capacity channel whose marginal benefit is largest on the weaker tasks. **Third**, the high-relevance group still shows the largest absolute gain (+2.1 at b=10), so the bias helps even tasks the network already handles well, but the increment shrinks relative to the medium group. Together these point to a *structured* capacity effect: the bias lifts all in-domain tasks above chance, but the lift is largest where the network's own routing is weakest — and it does nothing for tasks with no learnable structure. Together these rule out task relevance as the mediating variable — but they do not license the opposite reading, that the intervention works *through* routing. The context-swap probe of Sec. 5.3 shows the repaired gate is context-degenerate, so the lift is delivered by a gate that no longer routes by context. What the dose–response establishes is narrower and still useful: the bias behaves like a structured capacity channel whose marginal benefit is largest where the network's own routing is weakest, and it delivers nothing where there is no structure to amplify.

5.5. Signed drift: no pushback at the parameter level

To test whether the saturation above reflects the learner undoing the bias, we re-ran the T=10 MNIST probe with per-phase signed instrumentation of the context-t gating column (own-segment entries), under b=0 and b=10 (n=3 seeds). Adversarial compensation predicts a systematic *negative* displacement of the own gating column in the bias-imposed runs. The

measurement shows the opposite. Under b=10 the own-segment entries drift *positive* in all ten phases (one-sample t over seeds: 223, 37, 17, 17, 8, 11, 9, 5.1, 4.4, 4.5; per-phase mean displacement decaying from +0.027 to +0.004), versus $\leq +0.004$ under b=0, significant only in the first two phases. The raw own gate stays within $|u| \leq 0.32$ across all seeds and phases (seed means within ± 0.16) — thirty times below the imposed b=10 and three times below even b=1 — so the offset remains fully in effect throughout training. The network absorbs the intervention instead of opposing it: task training reinforces the forced own-segment routing slightly, and the ceiling of the measured win rate is a property of the raw-gate scale, not of active compensation. A complementary probe asks whether the network would learn the bias on its own; it acquires only a small positive own-segment bias at the raw-gate scale, an order of magnitude below the smallest effective dose (Sec. 7).

5.6. Why forgetting trades back at depth

At T=10 the bias reduces forgetting; at T=20 MNIST it mildly increases it. The probe's measurement explains the sign flip. With the bias active, a fixed minority of neurons routes through the own segment in *every* phase; over a longer sequence, later phases increasingly share this minority substrate, so late-task updates overlap earlier anchors on the same (now doubly used) neurons — ℓ rises because more neurons are available early, but the shared minority pays for it at depth. The saturation of p_{own} (Sec. 5.3) caps how much substrate the repair can recruit; at T=20 the cap binds.

6. Related work

Active-dendrite networks.. Single-neuron compartmental models [1, 2, 3, 10] motivate multi-segment artificial neurons; Iyer et al. [4] operationalize context-gated segment competition for lifelong classification and report reduced interference, and the GWC route-suppression variant [11, 12] and neuromorphic implementations [13] extend the family. These works validate the architecture; to our knowledge none measures whether the intended segment–task specialization actually obtains in the trained model, which is the object of our audit — nor, consequently, the dead-neuron artifact that naive dominance statistics incur under production sampling budgets.

Elastic weight consolidation and its localizations.. EWC [6] protects parameters with a single global anchor; SI [14], MAS [15], and online variants [16] refine the importance estimate; masking approaches [17, 18, 19] hard-allocate capacity. DeLP [5] localizes EWC per neuron via usage, and our companion work diagnoses the anchor-accumulation problem and bounds the anchor stack [5]. The present study is orthogonal: it asks whether protection granularity should descend *below* the neuron and answers negatively, then explains the negative via a routing audit and repairs the routing instead.

Negative results and measurement validity.. Benchmark pitfalls [20, 21, 22, 23, 24] established that evaluation practice can manufacture effects; the dead-neuron artifact here is an instance inside the model rather than the benchmark: the instrumentation, not the metric, fabricates the signal.

7. Limitations

1. **Scale.** All results are on one-hidden-layer MLPs with permuted benchmarks and $T \leq 20$. The artifact and anti-specialization findings concern the interaction of kWTA mortality with argmax instrumentation and should be re-checked wherever segments exist (e.g. convolutional or transformer backbones with routing modules); the causal repair's transfer is untested beyond this family. The task-relevance probe (Sec. 5.4) shows the repair is a structured capacity effect rather than a routing prior: it lifts in-domain tasks above chance and compresses task-to-task variance, but adds nothing for zero-relevance (random-label) tasks — so its benefit is contingent on the task stream containing learnable structure, a condition that should be verified per architecture and benchmark before generalizing.
2. **The repair is a prior, and the prior bypasses the gate.** The constant bias encodes the correct assumption (own segment first) that the network fails to learn at effective strength. We measured this directly: given a trainable per-(task, segment) bias table initialized at zero, the network acquires only a small positive own-segment bias ($+0.113 \pm 0.031$ MNIST, $+0.083 \pm 0.025$ Fashion; n=7 runs each, $t(6) \geq 8.7$) — the raw-gate scale, an order of magnitude below the smallest effective dose — and its accuracy matches the unbiased baseline. A companion study (S4, [9]) then establishes what the bias actually does: it saturates the own-segment sigmoid until the gate stops depending on the context (context-swap flip rate exactly 0.0% at b=10), and on MNIST a gate-removal control reproduces the entire gain. The

repair therefore does not restore segment–task routing; it removes the network's dependence on it. On Fashion-MNIST the bypass is rejected (66.84 vs. 64.30, t=6.8), so the mechanism is dataset-dependent and this paper's causal claim should be read as "a constant own-segment bias improves the trade-off", with the routing-restoration reading withdrawn. We also tested scheduled variants (Suppl. S4): an annealed bias ($b_p = 10(1 - p/T)$) retains 91–103% of the constant-bias learning gain at T=20 but does not reliably remove the BWT cost (significant improvement in one of two datasets only), while a ramped bias ($b_p = 10p/(T-1)$) *exceeds* the constant gain everywhere (123–197% retention, $t \geq 2.6$) at an equally or more negative BWT cost. Scheduling moves along the retention–plasticity frontier — tracking cumulative bias exposure, not timing — but does not move the frontier itself; separating gain from cost appears to require a mechanism (e.g. task-staggered protection), not a schedule.

3. **Depth-dependent forgetting trade-off.** At T=20 the BWT cost of the bias is small but real: 3/10 cells significantly negative (-0.66 to -0.95), the rest indistinguishable from baseline. We report it rather than select it away; the identity Eq. (1) makes explicit that accuracy gains can coexist with mildly worse retention.
4. **Granularity null is architecture-bound.** The plain→neuron step's dominance may weaken in architectures where segment parameters are genuinely disjoint (no shared feed-forward weights); our null is about this family's shared-substrate design.

8. Conclusion

We asked whether the segment structure of active-dendrite networks offers protection granularity below the neuron, and found that the question dissolves one level up: the entire benefit of usage-localized protection is realized at neuron level because the segment–task specialization the finer granularity would need is absent — measured, at chance before each phase and pushed below chance by each phase's own training, with Fisher mass too diffuse to partition meaningfully. On the way we isolated a dead-neuron artifact that fabricates dominance statistics for any naive instrument in this family, and showed that a one-parameter causal repair of the own-segment input raises the learning term in every tested cell (significant in 16/18 dose cells). The network itself reveals — through a dose-invariant saturated minority flip — that the repair is cheap because it need only saturate a few neurons; a companion study (S4, [9]) then shows *what* saturating them achieves: not restored routing, but a gate made context-degenerate, whose benefit a gate-removal control reproduces on MNIST. The honest summary is that we have measured the size and the cost of the repair precisely, and located its mechanism only far enough to know it is not the one we first assumed. Auditing the assumption before building on it took one paper's worth of null results; we hope the corrected instrumentation and the repair make the next one shorter.

Data availability

All code, per-run records (JSON), and analysis scripts are available at `https://github.com/zhouranxi/delp-anchor-budget` (experiments `s1_*`,

gevo_*, c2*; figures via paper_s1/make_figs.py).

Declaration of competing interest

The author declares no competing interests.

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Table 1: Granularity ladder (unbounded anchor stacks, n=10 paired seeds). ℓ/avg/BWT are absolute means; the right block is the paired difference vs. the neuron-resolved arm. All plain−neuron differences are significant at $|t| > 10$. For branch vs. neuron the learning term is indistinguishable in all six cells (largest $|t| = 1.3$ on $\Delta\ell$); the only significant branch−neuron differences are marginal forgetting effects (largest $|t| = 4.0$, ΔBWT ≤ 0.43 points), which we do not read as a granularity effect.

		absolute means			paired vs. neuron		
cell	arm	ℓ	avg	BWT	$\Delta\ell$	Δavg	ΔBWT
Permuted-MNIST, λ=1000							
T=10	plain	69.63	61.25	−9.31	−4.66*	−2.61*	+2.26*
T=10	neuron	74.28	63.86	−11.58	—	—	—
T=10	branch	74.21	64.13	−11.20	−0.08	+0.27	+0.38*
T=20	plain	46.35	37.30	−9.53	−5.44*	−1.35*	+4.31*
T=20	neuron	51.79	38.64	−13.84	—	—	—
T=20	branch	51.98	39.03	−13.63	+0.19	+0.39*	+0.21
Permuted-MNIST, λ=5000							
T=10	plain	54.41	51.69	−3.03	−4.78*	−2.80*	+2.19*
T=10	neuron	59.19	54.49	−5.22	—	—	—
T=10	branch	59.21	54.71	−5.01	+0.03	+0.22	+0.21*
T=20	plain	33.97	31.68	−2.40	−3.72*	−1.74*	+2.08*
T=20	neuron	37.68	33.42	−4.49	—	—	—
T=20	branch	37.59	33.39	−4.42	−0.09	−0.03	+0.07
Permuted-Fashion, λ=5000							
T=10	plain	38.53	33.47	−5.62	−3.76*	−1.65*	+2.35*
T=10	neuron	42.29	35.12	−7.96	—	—	—
T=10	branch	42.05	35.27	−7.53	−0.24	+0.15	+0.43
T=20	plain	26.56	22.75	−4.01	−2.78*	−1.22*	+1.65*
T=20	neuron	29.35	23.97	−5.66	—	—	—
T=20	branch	29.35	24.14	−5.49	+0.01	+0.17	+0.17

Table 2: Single-anchor references (Permuted-MNIST, n=10). ℓ is the learning term of Eq. (1) (mean over tasks of the accuracy right after each task is learned); ℓ_{final} is the accuracy on the last task alone and is reported separately because for these two arms it differs substantially from ℓ — SI in particular barely learns the later tasks at all, which is visible only in ℓ_{final}.

	method (λ)	ℓ	ℓ_{final}	avg	BWT
T=10	SI (100)	40.01	13.23	39.26	−0.83
T=10	MAS (1)	85.71	87.00	66.61	−21.21
T=20	SI (100)	25.80	11.19	25.14	−0.69
T=20	MAS (1)	86.69	87.91	55.75	−32.56

Table 3: Granularity under FIFO anchor budgets (λ=1000, n=10 paired). No branch−neuron difference is significant positive in any of the eight cells.

cell	M	neuron ℓ	branch ℓ	$\Delta\ell$ (t)	ΔBWT (t)
MNIST T=10	2	83.06	83.01	−0.05 (−1.0)	+0.22 (+1.1)
MNIST T=10	5	77.09	77.02	−0.07 (−0.5)	+0.15 (+1.0)
MNIST T=20	2	80.23	80.32	+0.09 (+1.0)	+0.39 (+1.9)
MNIST T=20	5	68.30	68.19	−0.11 (−0.8)	+0.28 (+1.5)
Fashion T=10	2	66.66	66.33	−0.33 (−2.3)*	−0.41 (−1.2)
Fashion T=10	5	58.90	58.74	−0.16 (−0.9)	+0.43 (+1.2)
Fashion T=20	2	62.61	62.72	+0.10 (+0.4)	−0.18 (−0.3)
Fashion T=20	5	49.79	50.23	+0.44 (+1.5)	−0.33 (−0.7)

Table 4: Causal probe, T=10, λ=1000: paired differences vs. b=0 (n=10; *: $|t| > 2$). p_{own} is the own-segment win rate over active neurons, computed per phase and averaged over phases. The two T=20 branch cells at b=1 were not collected.

dataset	arm (b)	$\Delta\ell$	Δavg	ΔBWT	p_{own}
MNIST	neuron (1)	$+1.99^*$	$+3.40^*$	$+1.56^*$	0.19
MNIST	neuron (3)	$+2.68^*$	$+4.08^*$	$+1.56^*$	0.17
MNIST	neuron (10)	$+3.32^*$	$+4.80^*$	$+1.65^*$	0.16
MNIST	branch (1)	$+2.18^*$	$+3.29^*$	$+1.23^*$	0.19
MNIST	branch (3)	$+2.90^*$	$+4.30^*$	$+1.57^*$	0.17
MNIST	branch (10)	$+3.16^*$	$+4.17^*$	$+1.13^*$	0.16
Fashion	neuron (1)	$+0.87$	$+1.03$	$+0.18$	0.15
Fashion	neuron (3)	$+1.23^*$	$+2.56^*$	$+1.48^*$	0.14
Fashion	neuron (10)	$+1.39^*$	$+0.69$	-0.77	0.13
Fashion	branch (1)	$+0.94^*$	$+2.08^*$	$+1.26$	0.15
Fashion	branch (3)	$+0.81$	$+1.79^*$	$+1.09^*$	0.14
Fashion	branch (10)	$+0.97^*$	$+1.03^*$	$+0.06$	0.13

Table 5: Causal probe, T=20: paired differences vs. b=0 (n=10; *: $|t| > 2$). p_{own} is the own-segment win rate over active neurons, computed per phase and averaged over phases. The two T=20 branch cells at b=1 were not collected.

dataset	arm (b)	$\Delta\ell$	Δavg	ΔBWT	p_{own}
MNIST	neuron (1)	$+2.14^*$	$+2.78^*$	$+0.68^*$	0.14
MNIST	neuron (3)	$+3.72^*$	$+3.25^*$	-0.49	0.14
MNIST	neuron (10)	$+4.56^*$	$+3.81^*$	-0.79^*	0.13
MNIST	branch (3)	$+3.66^*$	$+3.03^*$	-0.66^*	0.14
MNIST	branch (10)	$+4.13^*$	$+3.48^*$	-0.68	0.13
Fashion	neuron (1)	$+1.78^*$	$+1.58^*$	-0.21	0.14
Fashion	neuron (3)	$+2.19^*$	$+2.09^*$	-0.11	0.14
Fashion	neuron (10)	$+1.94^*$	$+1.03^*$	-0.95^*	0.13
Fashion	branch (3)	$+1.44^*$	$+1.98^*$	$+0.57$	0.13
Fashion	branch (10)	$+2.18^*$	$+1.96^*$	-0.24	0.13